

# HTML & CSS Starter Guide

A hands-on guide for building your first personal website

Based on a real portfolio project  
(GitHub Pages + index.html)

# Contents

1. The Big Picture: HTML vs CSS
2. How Tags, Elements, and Attributes Work
3. What Is a Class?
4. Every Page Skeleton
5. Essential HTML Tags (with examples)
6. CSS Basics You Will Actually Use
7. Build Your Page Step by Step
8. Real Example: Your Portfolio Structure
9. Common Mistakes & Fixes
10. Quick Reference Cheat Sheet
11. Practice Exercises

## 1. The Big Picture: HTML vs CSS

Every webpage you see is built from two languages working together:

| Language | Job                                                | Analogy                                       |
|----------|----------------------------------------------------|-----------------------------------------------|
| HTML     | Structure and meaning — <i>what</i> is on the page | Skeleton / outline of a document              |
| CSS      | Appearance — <i>how</i> it looks                   | Paint, spacing, layout, photos as backgrounds |

Browser reads HTML → builds the page structure ↓ Browser reads CSS → applies colors, sizes, layout ↓ You see the finished website

**Remember in one sentence:** HTML says *what*; CSS says *how it looks*.

## 2. How Tags, Elements, and Attributes Work

### Tag

The label in angle brackets: `<p>`, `<h1>`, `<a>`.

### Element

The whole piece: opening tag + content + closing tag.

```
<h1>Khushi Gupta</h1>
  ↑ open   ↑ text   ↑ close
```

This entire thing is one `<h1>` element.

### Self-closing tags

Some tags have no closing tag and no inner content:

```
<br>

```

## Attributes

Extra information inside the opening tag:

```
<a href="https://github.com/username">GitHub</a>
  ↑ attribute name = "value"
```

| Attribute | Used on     | Purpose                       |
|-----------|-------------|-------------------------------|
| href      | <a>         | Where a link goes             |
| src       | <img>       | Path to an image file         |
| alt       | <img>       | Description of the image      |
| class     | Any element | A custom name for CSS styling |
| lang      | <html>      | Page language (e.g. en)       |

## 3. What Is a Class?

A **class** is a nickname you invent. It connects HTML to CSS.

```
<!-- HTML: label the element -->
<header class="banner"></header>
<main class="content">...</main>

/* CSS: style by that nickname (note the dot) */
.banner { height: 13rem; }
.content { max-width: 40rem; }
```

**Tag** = built-in type (header, p, a).

**Class** = your custom label (banner, content, card).

You can name a class anything: card, profile-box, links-section. The browser does not care about the name — only your CSS does.

## 4. Every Page Skeleton

Every HTML file follows this structure:

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Your Name</title>
  <style>
    /* CSS rules go here */
  </style>
</head>
<body>
  <!-- Everything visible goes here -->
</body>
</html>

```

| Part            | Visible? | Purpose                            |
|-----------------|----------|------------------------------------|
| <!DOCTYPE html> | No       | Tells browser this is modern HTML5 |
| <head>          | No       | Setup: title, charset, CSS         |
| <title>         | Tab only | Browser tab + search results       |
| <body>          | Yes      | All visible content                |

## 5. Essential HTML Tags (with examples)

### Headings: <h1> through <h6>

Largest to smallest. Use one <h1> per page (your name). Sections use <h2>.

```

<h1>Khushi Gupta</h1>
<h2>About</h2>
<h2>Links</h2>

```

### Paragraph: <p>

```

<p>Hi! I'm Khushi – Working on AI, Robotics, and Software Engineering.</p>

```

### Line break: <br>

Starts a new line *inside* the same paragraph. No closing tag.

```
<p>CS Student @ Purdue University <br>
  Class of 2028</p>
```

## Link: <a>

```
<a href="https://github.com/your-username">GitHub</a>
```

| href value           | Meaning                     |
|----------------------|-----------------------------|
| https://...          | External website            |
| about.html           | Another page in your folder |
| mailto:you@email.com | Opens email app             |
| #                    | Placeholder (replace later) |

## Bullet list: <ul> + <li>

```
<ul>
  <li><a href="https://github.com/...">GitHub</a></li>
  <li><a href="#">LinkedIn</a></li>
</ul>
```

## Numbered list: <ol> + <li>

```
<ol>
  <li>Learn HTML basics</li>
  <li>Build index.html</li>
  <li>Push to GitHub Pages</li>
</ol>
```

## Image: <img>

```

```

Put the image file in the same folder as index.html, or use a path like images/photo.jpg.

## Section tags (grouping content)

| Tag       | Use for                                  |
|-----------|------------------------------------------|
| <header>  | Top of page (banner area)                |
| <main>    | Primary page content (one per page)      |
| <section> | Themed block (About, Projects, Links)    |
| <footer>  | Bottom of page (copyright, contact)      |
| <nav>     | Navigation menu                          |
| <div>     | Generic container when nothing else fits |

## Nesting rule

Tags open and close like nested boxes — last opened, first closed:

```
<main>
  <ul>
    <li><a href="...">Link</a></li>
  </ul>
</main>
```

## 6. CSS Basics You Will Actually Use

### Where CSS lives

```
<style>
  body { margin: 0; }
</style>
```

### Rule structure

```
selector {
  property: value;
  property: value;
}
```

## Selectors

| CSS     | Matches                         |
|---------|---------------------------------|
| body    | The <body> tag                  |
| .banner | Any element with class="banner" |
| h1      | All <h1> elements               |

## Properties from your real site

```
/* Center content column with side space */
.content {
  max-width: 40rem;      /* don't get wider than ~640px */
  margin: 0 auto;        /* center horizontally */
  padding: 2rem 6rem;     /* space inside the box */
}

/* LinkedIn-style banner header */
.banner {
  height: 13rem;         /* tall strip, measured top to bottom */
  background-image: url("photo.jpg");
  background-size: cover; /* fill area, crop if needed */
  background-position: center; /* show middle of photo */
}
```

## Units: rem

rem = relative to the root font size (usually 16px).

- 1rem  $\approx$  16px
- 2rem  $\approx$  32px
- 13rem  $\approx$  208px banner height

height on .banner is measured from the **top** of the banner to the **bottom** — not from the center of the screen.



## Image in page vs background image

|              | <code>&lt;img src="..."&gt;</code> | <code>background-image: url(...)</code>  |
|--------------|------------------------------------|------------------------------------------|
| Best for     | Profile photos, content images     | Banners, decorative headers              |
| Path syntax  | <code>src="file.jpg"</code>        | <code>url("file.jpg")</code>             |
| Size control | <code>width, max-width</code>      | <code>background-size, box height</code> |

## 7. Build Your Page Step by Step

1. **Create** `index.html` with the page skeleton (Section 4).
2. **Add your name** in `<h1>` and a short bio in `<p>`.
3. **Add sections** with `<h2>` for About, Links, Projects.
4. **Add links** with `<ul>` and `<a href="...">`.
5. **Add CSS** in `<style>` to center content (`max-width, margin: 0 auto`).
6. **Add a banner**: empty `<header class="banner">` + background CSS.
7. **Save and preview**: double-click the file or open via a local server.
8. **Push to GitHub** for your live site at `username.github.io`.

## 8. Real Example: Your Portfolio Structure

```
<header class="banner" aria-label="Profile header"></header>

<main class="content">
  <h1>Khushi Gupta</h1>
  <p>CS Student @ Purdue University <br>
    Class of 2028</p>

  <h2>About</h2>
  <p>Hi! I'm Khushi – Working on AI, Robotics, and Software Engineering.</p>

  <h2>Links</h2>
  <ul>
    <li><a href="https://github.com/your-username">GitHub</a></li>
    <li><a href="#">Another link</a></li>
  </ul>
</main>
```

Full-width banner (photo as background) | `.banner`  
 (13rem tall) | | Centered text column  
 | | `.content` | | `h1, p, h2, links` |

## 9. Common Mistakes & Fixes

| Problem                       | Likely cause                      | Fix                                                                                                         |
|-------------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------|
| Broken image icon             | Wrong filename or path            | Match <code>src</code> or <code>url()</code> to exact file name                                             |
| CSS not working               | Typo in class name                | <code>class="banner"</code> must match <code>.banner</code>                                                 |
| Link goes nowhere             | <code>href="#"</code> placeholder | Replace with real URL                                                                                       |
| Wrong part of photo in banner | Default crop                      | Try <code>background-position: center top;</code>                                                           |
| Page stuck at left edge       | No centering CSS                  | Add <code>max-width + margin: 0 auto</code>                                                                 |
| Tags look wrong               | Unclosed tag                      | Every <code>&lt;tag&gt;</code> needs <code>&lt;/tag&gt;</code> (except <code>br</code> , <code>img</code> ) |

## 10. Quick Reference Cheat Sheet

| I want to...               | HTML                                                       |
|----------------------------|------------------------------------------------------------|
| Page title in browser tab  | <code>&lt;title&gt;Name&lt;/title&gt;</code> in head       |
| Main page title            | <code>&lt;h1&gt;...&lt;/h1&gt;</code>                      |
| Section title              | <code>&lt;h2&gt;...&lt;/h2&gt;</code>                      |
| Paragraph                  | <code>&lt;p&gt;...&lt;/p&gt;</code>                        |
| Line break                 | <code>&lt;br&gt;</code>                                    |
| Clickable link             | <code>&lt;a href="URL"&gt;text&lt;/a&gt;</code>            |
| Bullet list                | <code>&lt;ul&gt;&lt;li&gt;...&lt;/li&gt;&lt;/ul&gt;</code> |
| Photo in page              | <code>&lt;img src="..." alt="..."&gt;</code>               |
| Photo as header background | CSS <code>background-image: url("...");</code>             |
| Center narrow column       | <code>max-width: 40rem; margin: 0 auto;</code>             |

## 11. Practice Exercises

### Exercise 1 — Add a Projects section

Inside `<main>`, add an `<h2>Projects</h2>`, a `<p>` description, and a `<ul>` with one link to a GitHub repo.

### Exercise 2 — Fix your GitHub link

Replace `your-username` in the GitHub URL with your real GitHub username. Save and click the link to test.

### Exercise 3 — Adjust the banner

Change `height: 13rem` to `18rem`. Refresh. Then try `background-position: center top`; and compare.

### Exercise 4 — Create a "card" class

Wrap your About section in `<section class="card">`. In CSS add: `.card { padding: 1rem; border: 1px solid #ccc; margin-bottom: 1rem; }`

### Exercise 5 — Add a footer

After `</main>`, add `<footer class="content"><p>© 2026 Khushi Gupta</p></footer>`

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Keep this guide next to your `index.html` while you edit.

Save → refresh browser → repeat.